

## What the Leftover Hash Lemma Gives

*And What It Cannot: a Limitation on the Collision Route*

**Resolves.** Nothing. This is a companion note to *Randomness Extraction from Unpredictable Random-Oracle Sources: A Leftover-Hash-Lemma Conjecture, Public Seed* (`main.tex` in this folder), whose conjecture is proved in `proof.tex` beside it. What is recorded here is one calibration (Proposition 2) and one limitation (Theorem 1): the leftover hash lemma matches the conjectured bound on its first summand exactly, and on its second it falls short of it by a factor of order  $\sqrt{W/\log_2 D}$ , where  $W \approx \min\{R, D\epsilon\}$  is defined precisely in Theorem 1. Section 5 says what is *not* claimed.

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**Date:** 29 August 2026

## 1 Setting

The setting is that of `main.tex`, repeated here so this note stands alone. Fix nonempty finite sets  $\mathcal{K}$  (seeds),  $\mathcal{D}$  (inputs),  $\mathcal{R}$  (outputs), and write  $K := |\mathcal{K}|$ ,  $D := |\mathcal{D}|$ ,  $R := |\mathcal{R}|$ . A table  $H$  is drawn uniformly from  $\text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ . A source  $S$ , given the *entire* table, outputs a pair  $(x, z) \in \mathcal{D} \times \{0, 1\}^*$ ; a seed  $sd \xleftarrow{\$} \mathcal{K}$  is drawn *after*  $S$  has terminated; and an unbounded distinguisher, given  $(H, sd, y_b, z)$ , is to tell  $y_0 = H(sd, x)$  from a uniform  $y_1$ . Write  $\text{Adv}^{\text{ext-pub}}(S, D)$  for twice its winning probability minus one. The source is  $\epsilon$ -unpredictable if no unbounded predictor given  $(H, z)$  guesses  $x$  with probability more than  $\epsilon$ . Conjecture 1 of `main.tex` asserts a universal constant  $c$  with

$$\text{Adv}^{\text{ext-pub}}(S, D) \leq c \left( \sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}} \right), \quad (1)$$

and `proof.tex` proves it with  $c = 2$ .

Throughout, condition on a table and an auxiliary output,  $(H, Z) = (h, z)$ , and write  $p_x := \Pr[X = x \mid h, z]$  for the source's conditional law,  $\epsilon_{h,z} := \max_x p_x$ , and

$$\text{CP} := \sum_{x \in \mathcal{D}} p_x^2, \quad c_k := \Pr_{X, X'}[h(k, X) = h(k, X')], \quad \bar{c} := \frac{1}{K} \sum_{k \in \mathcal{K}} c_k,$$

where  $X, X'$  are independent draws from  $p$ . Note  $\text{CP} \leq \epsilon_{h,z}$ , since  $\sum_x p_x^2 \leq (\max_x p_x) \sum_x p_x$ . Two elementary facts are used repeatedly and are recorded once.

**Fact 1 (Cauchy–Schwarz, two forms).** Let  $q$  be a probability distribution on a finite set  $\Omega$  and let  $A \subseteq \Omega$  carry all of its mass. Then

$$\sum_{\omega} q_{\omega}^2 \geq \frac{1}{|A|}. \quad (2)$$

If moreover  $u$  is uniform on  $\Omega$ , then

$$\text{SD}(q, u) = \frac{1}{2} \|q - u\|_1 \leq \frac{1}{2} \sqrt{|\Omega|} \|q - u\|_2 = \frac{1}{2} \sqrt{|\Omega| \sum_{\omega} q_{\omega}^2 - 1}. \quad (3)$$

*Proof.* For (2),  $1 = (\sum_{\omega \in A} q_{\omega})^2 \leq |A| \sum_{\omega \in A} q_{\omega}^2 \leq |A| \sum_{\omega} q_{\omega}^2$ . For (3), the inequality is  $\|v\|_1 \leq \sqrt{|\Omega|} \|v\|_2$  applied to  $v = q - u$ , and the final equality is  $\|q - u\|_2^2 = \sum_{\omega} q_{\omega}^2 - 1/|\Omega|$ , which follows by expanding the square and using  $\sum_{\omega} q_{\omega} = 1$ .  $\square$

## 2 The leftover-hash step

**Proposition 1 (The bound the lemma returns).** *For every table  $h$  and every view  $z$  of positive probability,*

$$\max_{\mathcal{D}} (2 \Pr[\mathcal{D} \text{ wins} \mid h, z] - 1) \leq \frac{1}{2} \sqrt{R \bar{c} - 1},$$

and consequently  $\text{Adv}^{\text{ext-pub}}(S, \mathcal{D}) \leq \mathbb{E}_{(H, Z)} \left[ \frac{1}{2} \sqrt{R \bar{c} - 1} \right]$ .

*Proof.* Because  $sd$  is drawn only after  $S$  has terminated, it is uniform on  $\mathcal{K}$  and independent of  $X$  given  $(h, z)$ . The distinguisher's view therefore differs between the two challenge bits only in the pair  $(sd, y_b)$ , which is distributed as  $Q := \text{law}(sd, h(sd, X))$  under  $b = 0$  and as the uniform distribution  $U$  on  $\mathcal{K} \times \mathcal{R}$  under  $b = 1$ . The optimal advantage of an unbounded distinguisher between two distributions is their statistical distance, so the left-hand side is  $\text{SD}(Q, U)$ .

Now  $|\mathcal{K} \times \mathcal{R}| = KR$  and

$$\sum_{k, r} Q(k, r)^2 = \Pr[(sd, h(sd, X)) = (sd', h(sd', X'))] = \sum_k \frac{1}{K^2} c_k = \frac{\bar{c}}{K},$$

the middle equality because  $sd, sd'$  are independent and uniform, so both equal a given  $k$  with probability  $1/K^2$ , and given  $sd = sd' = k$  the outputs collide with probability  $c_k$ . Feeding this into (3) gives  $\text{SD}(Q, U) \leq \frac{1}{2} \sqrt{KR \cdot \bar{c}/K - 1} = \frac{1}{2} \sqrt{R \bar{c} - 1}$ . Averaging over  $(H, Z)$  gives the second claim.  $\square$

Writing  $c_k = \text{CP} + \beta_k$  with  $\beta_k := \sum_{x \neq x'} p_x p_{x'} \mathbf{1}[h(k, x) = h(k, x')]$  and  $\bar{\beta} := \frac{1}{K} \sum_k \beta_k$  splits the quantity under the root as

$$R \bar{c} - 1 = R \cdot \text{CP} + \gamma, \quad \gamma := R \bar{\beta} - 1, \quad (4)$$

the first summand carrying the source's entropy deficiency and the second the family's failure to be universal on the source's support.

## 3 Calibration: the lemma is sharp on the first summand

**Proposition 2 (Universal families).** *Suppose the family  $\{h(k, \cdot)\}_{k \in \mathcal{K}}$  is 2-universal, that is  $\Pr_k[h(k, x) = h(k, x')] \leq 1/R$  for all  $x \neq x'$ . Then  $\gamma \leq 0$  and, for every  $\epsilon$ -unpredictable source,*

$$\text{Adv}^{\text{ext-pub}}(S, \mathcal{D}) \leq \frac{1}{2} \sqrt{\epsilon R}.$$

*Proof.* Exchanging the order of summation,  $\bar{\beta} = \sum_{x \neq x'} p_x p_{x'} \Pr_k[h(k, x) = h(k, x')] \leq (1 - \text{CP})/R$ , so  $\gamma \leq 1 - \text{CP} - 1 = -\text{CP} \leq 0$ . By (4) and Proposition 1 the conditional bound is at most  $\frac{1}{2}\sqrt{R \cdot \text{CP}} \leq \frac{1}{2}\sqrt{R \epsilon_{h,z}}$ , using  $\text{CP} \leq \epsilon_{h,z}$ . Averaging over  $(H, Z)$  and applying Jensen's inequality to the concave nondecreasing  $\sqrt{\cdot}$ , together with  $\mathbb{E}[\epsilon_{H,Z}] \leq \epsilon$ , gives the claim.  $\square$

So on the first summand of (1) the leftover hash lemma loses nothing: it delivers that summand with  $c = \frac{1}{2}$ , and the proof in proof.tex delivers it with  $c = 1/\sqrt{2}$ , the difference being only that note's passage from  $\epsilon_{h,z}$  to the integer  $\lfloor 1/\epsilon_{h,z} \rfloor$ . Everything that separates the two routes lies in  $\gamma$ .

## 4 The limitation

Proposition 2 does not apply here, and cannot be made to. A 2-universal family on  $D$  inputs needs  $\Omega(\log D)$  bits of seed, that is  $K \geq D^{\Omega(1)}$ , whereas (1) is of interest at  $K \approx \log_2 D$ ; at that seed length no universal family exists at all, for any table. So  $\gamma$  must be taken over the supports the source may choose after reading  $H$ , and the theorem below says how large those make it. Recall that the source sees the entire table before  $sd$  exists, so any support it can name from  $H$  is admissible.

**Theorem 1 (The collision route cannot beat  $\sqrt{W/K}$ ).** *Let  $\mathcal{K}, \mathcal{D}, \mathcal{R}$  be nonempty and finite and let  $\epsilon \in [1/D, 1]$ . Put*

$$t := \lceil 1/\epsilon \rceil, \quad \sigma := \lceil Rt/D \rceil, \quad W := \lfloor R/\sigma \rfloor,$$

*so that  $1 \leq t \leq D$ ,  $1 \leq \sigma \leq R$  and  $1 \leq W \leq R$ . Then for every table  $h \in \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$  there is an  $\epsilon$ -unpredictable source  $S$  — uniform on a set of  $t$  inputs, with  $z = \perp$  — for which*

$$R \bar{c} - 1 \geq \frac{W - 1}{K}, \quad \text{so that the bound of Proposition 1 is at least } \frac{1}{2} \sqrt{\frac{W - 1}{K}}.$$

*No randomness is used: this holds for every table, not merely for most. When  $R$  and  $D/t$  are powers of two, as they are in the parameter ranges of interest,  $W = \min\{R, D/t\}$  exactly; in general  $W \geq \frac{1}{2} \min\{R, D/t\} - 1$ .*

*Proof.* The parameters are in range. From  $\epsilon \geq 1/D$  we get  $t = \lceil 1/\epsilon \rceil \leq D$ , hence  $Rt/D \leq R$  and  $\sigma \leq R$ ; and  $\sigma \geq 1$ , so  $W = \lfloor R/\sigma \rfloor \geq 1$  and  $W \leq R$ . If  $W = 1$  the claimed inequality is vacuous, so assume  $W \geq 2$ .

*The support.* Fix any seed  $k_1 \in \mathcal{K}$  and let  $S \subseteq \mathcal{R}$  consist of  $\sigma$  symbols  $r$  maximising  $|\{x \in \mathcal{D} : h(k_1, x) = r\}|$ . The  $D$  inputs are distributed among the  $R$  symbols, so the  $\sigma$  most popular of them carry at least a  $\sigma/R$  fraction of the total, that is at least  $D\sigma/R$  inputs; and  $\sigma \geq Rt/D$  gives  $D\sigma/R \geq t$ . So  $\{x : h(k_1, x) \in S\}$  has at least  $t$  elements; let  $T$  be any  $t$  of them and let  $S(h)$  output  $x \xleftarrow{\$} T$  together with  $z \leftarrow \perp$ .

*Admissibility.*  $T$  is computed from  $h$ , which  $S$  receives in full, and  $sd$  is drawn only after  $S$  terminates, so  $S$  is a legitimate source for the game. Its conditional law is uniform on  $T$ , so  $\epsilon_{h,z} = 1/t = 1/\lceil 1/\epsilon \rceil \leq \epsilon$ , and  $S$  is  $\epsilon$ -unpredictable.

*The two collision bounds.* For every  $k$  the law of  $h(k, X)$  is a distribution on  $\mathcal{R}$ , so (2) with  $A = \mathcal{R}$  gives  $c_k \geq 1/R$ . For  $k = k_1$  every  $x \in T$  has  $h(k_1, x) \in S$ , so the law of  $h(k_1, X)$  is carried by  $S$  and (2) with  $A = S$  gives  $c_{k_1} \geq 1/\sigma$ .

*Assembling.* Averaging the  $K$  lower bounds and using  $R/\sigma \geq \lfloor R/\sigma \rfloor = W$ ,

$$R\bar{c} - 1 \geq R \left[ \frac{1}{K} \cdot \frac{1}{\sigma} + \frac{K-1}{K} \cdot \frac{1}{R} \right] - 1 = \frac{R/\sigma}{K} + \frac{K-1}{K} - 1 \geq \frac{W}{K} - \frac{1}{K} = \frac{W-1}{K},$$

and Proposition 1 applied to this source returns  $\frac{1}{2}\sqrt{R\bar{c} - 1} \geq \frac{1}{2}\sqrt{(W-1)/K}$ .

*The two forms of  $W$ .* If  $R = 2^m$  and  $D/t = 2^\ell$  then  $Rt/D = 2^{m-\ell}$ , so  $\sigma = \max\{1, 2^{m-\ell}\}$  and  $W = \lfloor R/\sigma \rfloor = \min\{2^m, 2^\ell\} = \min\{R, D/t\}$ . In general  $\sigma \leq Rt/D + 1$ , so  $R/\sigma \geq R/(Rt/D + 1) = \min\{R, D/t\} \cdot \frac{\max\{R, D/t\}}{Rt/D+1} \geq \frac{1}{2} \min\{R, D/t\}$ , using  $Rt/D + 1 \leq 2 \max\{Rt/D, 1\}$ ; hence  $W \geq \frac{1}{2} \min\{R, D/t\} - 1$ .  $\square$

**Remark 1 (Pinning more than one row).** Confining  $j$  rows rather than one, each to  $\sigma$  symbols, leaves at least  $D(\sigma/R)^j$  inputs and replaces  $W - 1$  by about  $j(W - 1)$ ; the constraint is that this stay at least  $t$ , so  $j \leq \log_W(D/t)$ . The gain is a factor  $\sqrt{j}$  in the bound, that is  $\log_2 j$  further bits of seed in the comparison below — one or two in the parameter ranges of interest. The one-row construction of Theorem 1 is therefore essentially the whole effect, and is what the statement records.

**Corollary 1 (The comparison, in seed length).** Fix a target error  $\delta \in (0, 1)$  and suppose the entropy deficiency is not the binding constraint, so that attention is on the second summand of (1). Driving that summand below  $\delta$  requires, by (1) with the constant  $c = \frac{6}{5}$  of proof.tex, a seed space of size  $K \geq \frac{36}{25}(1 + \ln D)/\delta^2$ ; whereas by Theorem 1 the leftover-hash route requires  $K \geq (W - 1)/(4\delta^2)$ . In bits,

$$\log_2 K = \log_2 \log_2 D + 2\log_2(1/\delta) + O(1) \quad \text{against} \quad \log_2 K = \log_2 W + 2\log_2(1/\delta) - 2.$$

Both are  $2\log_2(1/\delta)$  in the error; they differ in the first term, and there by  $\log_2 W$  against  $\log_2 \log_2 D$ .

The two bounds therefore have the same shape and the same rate in  $K$ , and differ in one symbol: what stands under the second square root. The conjecture pays  $\log_2 D$ , a number of bits; the lemma pays  $W$ , the size of a set. At  $D = 2^{256}$ ,  $R = 2^{64}$  and a source of min-entropy 128, reaching  $\delta = 2^{-32}$  needs  $\log_2 K \approx 72$  under (1) and 126 by this route.

**Remark 2 (Where the loss comes from, and when it vanishes).** The loss is the step (3),  $\|v\|_1 \leq \sqrt{KR} \|v\|_2$ , which is tight only when  $v$  is spread evenly over all  $KR$  cells. The source of Theorem 1 makes it as far from that as possible: its deviation is concentrated on one row out of  $K$ , where the step is off by about  $\sqrt{RK}$ . Indeed the true advantage against that source is  $O(1/K)$  — one row of the  $K$  is biased and the rest are not — while the lemma returns  $\frac{1}{2}\sqrt{(W-1)/K}$ .

The gap closes exactly when  $W$  does. At  $\epsilon = 1/D$  one has  $W = \min\{R, 1\} = 1$  and Theorem 1 is vacuous, correctly: a source of maximal entropy is uniform on all of  $\mathcal{D}$  and has no support to select. This is the same boundary proof.tex records when its own selection term  $\frac{1}{t} \ln \binom{D}{t}$  vanishes at  $t = D$ . More generally  $W$  is small whenever

the source's min-entropy is close to  $\log_2 D$  or the output is short, and in those regimes the route loses little or nothing: for  $D = 2^{256}$ ,  $R = 2^{64}$  and min-entropy 254 one has  $W = 4$  and Theorem 1 exhibits no loss at all.

## 5 What is not claimed

Theorem 1 is a limitation and only that: it says the collision route cannot return a bound better than  $\frac{1}{2}\sqrt{(W-1)/K}$  on its second summand. It does *not* say the route attains that, and this note does not prove a matching upper bound.

The gap is real and should not be papered over. Certifying  $\gamma$  for a random table by the same bounded-differences argument `proof.tex` uses —  $\tilde{\beta}$  has bounded-differences constant  $2/(Kt)$  in the  $Kt$  table entries above a support of size  $t$ , and a union bound over the  $\binom{D}{t}$  supports of every size costs  $e^{t(1+\ln D)}$  — yields only  $\gamma \leq R\sqrt{2(1+\ln D)/K}$  with high probability, hence an upper bound of  $\frac{1}{2}\sqrt{\epsilon R} + \frac{1}{2}\sqrt{R} (2(1+\ln D)/K)^{1/4}$ . That is larger than Theorem 1's floor by a factor of order  $(K \ln D)^{1/4}$ , so the truth for this route lies somewhere between, and locating it is not attempted here. What the two together do establish is that the route cannot reach (1): the floor alone already exceeds it by  $\log_2 W - \log_2 \log_2 D$  bits of seed.

Nor is anything here a statement about the conjecture. (1) is true — `proof.tex` proves it with  $c = 2$  — and what this note adds is only that it is not a corollary of the leftover hash lemma, and that its content is the second summand: the price of letting the source choose its support after seeing the entire table.